

Intelligent Healthcare Compliance and Safety Management System

Waterfall Project Management Methodology

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Project Duration: 7 Weeks | Methodology: Waterfall | Domain: Healthcare Compliance & Safety

01

Requirements Analysis

- Gather HIPAA/GDPR compliance needs
- Stakeholder interviews (hospitals, staff)
- Define safety reporting requirements
- Document functional & non-functional specs

Week 1

Deliverable: SRS Document

02

System Design

- Define system architecture (cloud + ML)
- Design DB schema for patient records
- UI/UX wireframes & data flow diagrams
- Security & compliance design (encryption)

Week 2

Deliverable: Design Document

03

Implementation / Development

- Build backend APIs & compliance engine
- Develop ML risk-prediction modules
- Implement frontend dashboards
- Integrate EHR systems & audit logging

Week 3 - 4

Deliverable: Working Codebase

04

Testing & Validation

- Unit, integration & system testing
- Compliance & security penetration tests
- User Acceptance Testing (UAT) with staff
- Performance & load benchmarking

Week 5

Deliverable: Test & QA Report

05

Deployment

- Production cloud deployment (HIPAA)
- User training & documentation rollout
- Phased go-live across hospital units
- Knowledge transfer & handover

Week 6

Deliverable: Live System

06

Maintenance & Support

- Continuous compliance monitoring
- Bug fixes, patches & security updates
- Model retraining & performance tuning
- Regulatory updates & audits

Week 7

Deliverable: Support & SLA

Project Overview

The Intelligent Healthcare Compliance and Safety Management System is designed to automate regulatory compliance (HIPAA, GDPR), incident reporting, and patient safety monitoring across healthcare facilities. It leverages machine learning for risk prediction and real-time alerts.

Key Principles of Waterfall

- ✓ Sequential, structured phases
- ✓ Each phase must complete before the next
- ✓ Comprehensive documentation at every stage
- ✓ Clear deliverables & milestones
- ✓ Ideal for regulated healthcare environments

Team Group W

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